

OFFICIAL ANSWER KEY & MARKING SCHEME

Class 12 Physics Practice Paper (2025-26)

Section A: MCQs (1 Mark Each)

1. (a) $1/x^2$

Reason: The electric potential V due to a short dipole at a distance x on its axis is given by $V = (1/4\pi\epsilon_0)(p/x^2)$.

2. (c) $2vd$

Reason: Drift velocity $vd = (e\tau/mL)V$. Thus, $vd \propto V$. If potential difference V is doubled, drift velocity doubles.

3. (c) They are always attractive.

Reason: This statement is NOT true. Nuclear forces become repulsive at very short distances (less than 0.8 fm) to prevent nuclear collapse.

4. (c) Microwaves

Reason: Microwaves are used in radar systems due to their short wavelength range, suitable for navigation.

5. (c) β/μ

Reason: Fringe width $\beta = \lambda D/d$. In liquid, $\lambda' = \lambda/\mu$. Therefore, the new fringe width is $\beta' = \beta/\mu$.

Section B: Very Short Answers (2 Marks)

6. Diffusion and Drift Current:

Diffusion: Concentration gradient (p to n holes, n to p electrons). Flows p-side to n-side.

Drift: Depletion region electric field sweeps minority carriers. Flows n-side to p-side.

7. Current Sensitivity:

Definition: Deflection per unit current. $I_s = \phi/I = NAB/k$.

Increase: Increase N, A, or B; or decrease torsional constant k.

8. Planck's Constant:

Formula: $v^2_{\max} = (2hc/m)(1/\lambda) - (2\phi_0/m)$.

Slope = $2hc/m \Rightarrow h = (m \times \text{Slope}) / 2c$. Intercept finds work function.

9. 1:1

Reason: Nuclear density is independent of mass number A. All nuclei have constant density $\approx 2.3 \times 10^{17} \text{ kg/m}^3$.

Section C: Short Answers & Derivations (3 Marks)

11. Huygens' Principle: Show plane wavefront incidence. Verify $\sin i / \sin r = v_1 / v_2 = \mu_{21}$.

13. LCR Circuit:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}.$$

$$\text{Resonance: } X_L = X_C \Rightarrow \omega_0 = 1/\sqrt{LC}.$$

15. BE Curve: Curve peaks at Fe-56. Fusion occurs for low A to increase BE; Fission for high A to split into more stable nuclei.

Section D: Case Study Answers (4 Marks)

16. Capacitors:

(i) $C = K\epsilon_0 A/d$ (Increases by factor K).

(ii) $V' = V/K = V/3$ (Since Q is constant).

(iii) **Charge** remains constant (Isolated capacitor).

17. Lenses:

- (i) $1/f = (1.5 - 1)(2/20) = 0.05 \Rightarrow f = 20 \text{ cm.}$
(ii) Focal length increases in water.

Section E: Long Answers (5 Marks)

18. Gauss Law:

Sheet: $E = \sigma/2\epsilon_0$. Parallel sheets: $E = \sigma/\epsilon_0$ (between) and $E = 0$ (outside).

19. Galvanometer:

Radial Field: Ensures $\sin \theta = 1$ always. Torque is maximum and constant. Scale becomes linear ($I \propto \phi$).

20. Cells & Numerical:

Eq. EMF: $E_{eq} = (E_1 r_2 + E_2 r_1)/(r_1 + r_2)$.
Numerical: $R = 17 \Omega$, $V = 8.5 \text{ V}$.

**Physics Last Paper Answer Sheet By Pragya Thunderstudy
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